

Lava Provider Guide

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Scope	User-facing guide to every provider Lava ships, its auth, its capabilities, and how downloads work

Every claim below is grounded in the real `*Descriptor.kt` and magnet-cache source files cited inline. Where a capability is declared on a descriptor but not yet exposed through the SDK, that gap is called out explicitly (no guessing — §11.4.6).

What a "provider" is in Lava

Lava is a multi-tracker client. Each supported site is a **provider** described by a `TrackerDescriptor` (the contract lives in `core/tracker/api/src/main/kotlin/lava/tracker/api/TrackerDescriptor.kt`). A descriptor declares:

- `trackerId / displayName` — the stable id and the label you see in the app.
- `baseUrls` — the primary site plus mirrors the SDK can fail over to.
- `capabilities` — the exact set of features the SDK will resolve for that provider. Per **Capability Honesty (constitutional clause §6.E)** a declared capability **MUST** map to a real, working feature — a provider never advertises something it cannot actually do.
- `authType` — how (or whether) you log in: `CAPTCHA_LOGIN`, `FORM_LOGIN`, or `NONE`.
- `verified` — whether the provider's end-to-end flow has passed a real-device Challenge Test (§6.G). `verified = false` means the provider is wired but its full user flow has a known gap (see the per-provider notes).

Capability matrix

Read directly from the six descriptor files. Legend:  = declared and resolved through the SDK;  = wired in code but **not** exposed via the descriptor (honest absence per §6.E); **✗** = not applicable to that provider.

Capability	RuTracker	RuTor	Kinozal	NNM-Club
Auth type	 CAPTCHA_LOGIN	 FORM_LOGIN	 FORM_LOGIN	 FORM_LOGIN
SEARCH				
BROWSE				
FORUM		✗	✗	not wired
TOPIC				
COMMENTS				
FAVORITES		✗	✗	no endpoint
TORRENT_DOWNLOAD (.torrent)				
MAGNET_LINK				
RSS	✗		✗	✗
AUTH_REQUIRED		 (optional)		
Anonymous browse/ search	✗		✗	✗
verified (Challenge- tested flow)	true	true	false	false
Text encoding	Windows-1251	UTF-8	windows-1251	windows-1251

Sources: RuTrackerDescriptor.kt, RuTorDescriptor.kt, KinozalDescriptor.kt, NnmclubDescriptor.kt, ArchiveOrgDescriptor.kt, GutenbergDescriptor.kt (all under core/tracker/<id>/src/main/kotlin/lava/tracker/<id>/).

Per-provider detail

RuTracker.org (rutracker)

- **What it is:** the largest Russian torrent tracker. Primary <https://rutracker.org>, with rutracker.net and rutracker.cr as failover mirrors.

- **Auth:** CAPTCHA_LOGIN — login can present a CAPTCHA you must solve. This is the only provider with CAPTCHA in its capability set.
- **What you can do:** search, browse forums (FORUM), open topics, read comments, keep favorites, and download both .torrent files and magnet links.
- **API support:** RuTracker is the one provider with a working route family in the lava-api-go backend today (apiSupported = true; see the RuTrackerDescriptor.kt Phase-1 hotfix note).
- **Verified:** yes — Challenge Tests C1, C2, C4, C5, C7, C8.
- **Note:** UPLOAD and USER_PROFILE were intentionally dropped from the descriptor because no SDK feature interface backs them (clause §6.E); the legacy upload/profile plumbing still ships but is not advertised.

RuTor.info (rutor)

- **What it is:** an anonymous-friendly Russian tracker. Mirrors include rutor.info, rutor.is, the www. variants, and an IPv6-only 6tor.org.
- **Auth:** FORM_LOGIN (a plain form POST — no CAPTCHA). AUTH_REQUIRED is declared so the SDK can expose optional actions like adding a comment, but **search and browse work without logging in** (supportsAnonymous = true).
- **What you can do:** search, browse categories (no nested forum tree — FORUM is intentionally absent), open topics, read/add comments, RSS, and download .torrent + magnet.
- **Verified:** yes — Challenge Tests C1, C3, C4, C6, C7, C8.

Kinozal.tv (kinozal)

- **What it is:** a Russian movie/media tracker. Mirrors kinozal.tv, kinozal.me.
- **Auth:** FORM_LOGIN.
- **What you can do:** search, browse, open topics, read comments, download .torrent + magnet.
- **Verified: no** (verified = false). The descriptor records a known post-login navigation gap on the onboarding screen (.lava-ci-evidence/sixth-law-incidents/2026-05-04-onboarding-navigation.json). Treat Kinozal as wired-but-not-yet-fully-verified end to end.

NNM-Club (nnmclub)

- **What it is:** a Russian tracker with an embedded forum. Mirrors nnmclub.to, nnm-club.me.
- **Auth:** FORM_LOGIN.
- **What you can do:** search, browse, open topics, read comments, download .torrent + magnet. **FORUM is not wired** through the SDK surface yet, and **FAVORITES has no scraping**

endpoint — both are honestly absent from the capability set (§6.E).

- **Verified: no** (verified = false) — same post-login navigation gap as Kinozal.

Internet Archive (archiveorg)

- **What it is:** a digital library, **not a torrent tracker** — archive.org, served over JSON + HTTP.
- **Auth:** NONE (implicitly anonymous).
- **What you can do:** search, browse, open topic metadata, and browse a forum surface. **No torrents and no magnets** — files are served over HTTP, so TORRENT_DOWNLOAD and MAGNET_LINK are deliberately absent.
- **Verified:** yes — Challenge Test C11 (continue → authorized main app).

Project Gutenberg (gutenberg)

- **What it is:** a free e-book library backed by the Gutendex JSON API (<https://gutendex.com>) — **not a torrent tracker**.
 - **Auth:** NONE (public, unauthenticated API).
 - **What you can do:** search, browse, open a book's topic page. It serves EPUB / plain-text / HTML e-books **over HTTP, not .torrent** — declaring TORRENT_DOWNLOAD would be a §6.E bluff, so it is absent.
 - **Verified:** yes — Challenge Test C12.
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How downloads work

Lava distinguishes two artifact kinds, by provider type.

Torrent providers (RuTracker, RuTor, Kinozal, NNM-Club)

These providers expose **two** download paths:

1. **.torrent file (TORRENT_DOWNLOAD).** Lava fetches the actual torrent file from the tracker's download endpoint and hands it off. The bytes are validated as a real bencoded torrent before use — see `core/common/src/main/kotlin/lava/common/torrent/TorrentFileValidator.kt` and `Bencode.kt`.
2. **Magnet link (MAGNET_LINK).** This is where the §6.E "Capability Honesty" design matters. The `DownloadableTracker.getMagnetLink` method is **synchronous**

(non-suspend), and its contract is to return the magnet only when it is available **without** an extra HTTP fetch — otherwise null (an honest absence, never a fabricated value).

To make that honest, Kinozal and NNM-Club use an in-memory, process-lifetime **magnet cache**:

- `core/tracker/kinozal/src/main/kotlin/lava/tracker/kinozal/magnet/KinozalMagnetCache.kt`
- `core/tracker/nmclub/src/main/kotlin/lava/tracker/nmclub/http/NmclubMagnetCache.kt`

The flow: when you **open a topic** (or, for NNM-Club, when a **search** result row is parsed) the production parser extracts the genuinely-present magnet and writes it into the cache (`put(topicId, magnet)`). The download feature later reads it back (`get(topicId)`). So **the magnet surfaces after you have viewed the topic/search** — at that point the synchronous lookup returns the real magnet with no hidden round-trip. If you ask for a magnet for a topic you have never opened, the cache returns null rather than inventing one. Both caches are `@Singleton`, backed by a thread-safe `ConcurrentHashMap`, so the topic feature and the download feature share one instance per process. Magnet links are validated by `core/common/.../torrent/MagnetLinkValidator.kt`.

Library providers (Internet Archive, Project Gutenberg)

These are **HTTP downloads, not torrents**. Internet Archive serves files directly; Project Gutenberg serves EPUB / plain-text / HTML e-books over HTTP via Gutendex. Neither declares `TORRENT_DOWNLOAD` or `MAGNET_LINK`. (Note from `GutenbergDescriptor.kt`: an HTTP-download implementation exists in code but is not exposed through `getFeature()` because `TrackerCapability` has no `HTTP_DOWNLOAD` value today — an honest not-yet-wired state.)

Quick reference: which providers need credentials?

Need to log in?	Providers
Yes (always)	RuTracker (CAPTCHA), Kinozal, NNM-Club
Optional (anonymous search/ browse works)	RuTor
No (anonymous)	

Need to log in?	Providers
	Internet Archive, Project Gutenberg

For how to enter and store credentials, see the in-app provider login flow; real credentials live only in the gitignored `.env` (\$6.H) and are never committed.